

Senior Rebuild Prompt: Focus Buddy Chrome Extension

Build a Manifest V3 Chrome extension called ****Focus Buddy****. It should be a cute, strict focus-session blocker with a retro desktop-buddy aesthetic. The extension must work without a backend or build step: plain HTML, CSS, and JavaScript files only.

Core Product

Create a browser extension that lets the user start a timed focus session with:

- * A text goal, such as "finish assignment" or "learn Next.js server actions".
- * Duration presets: 25 minutes, 50 minutes, and 90 minutes, with 50 minutes selected by default.
- * A user-editable allowlist of websites that are permitted during focus mode.
- * An optional Anthropic API key field for generating fresh, goal-aware motivational lines.
- * A visible focus streak count.
- * A small animated/cute buddy mascot that encourages the user while they work and scolds them when they open blocked sites.

During an active session, any 'http' or 'https' page whose hostname is not on the allowlist must be blocked. Do not merely redirect the tab; inject a full-page overlay into the current page so the user sees the blocked state immediately. The overlay must prevent scrolling, show the user goal, show a live countdown, show the current streak, and offer:

- * "Back to work" button, which calls 'history.back()' or navigates to 'about:blank' if there is no previous page.
- * "Notes" button, which opens a textarea for capturing distracting thoughts. Saved notes must be appended to 'chrome.storage.local.capturedNotes' and shown later in the popup.
- * "End session early" button, with confirmation. Ending early should not increment the streak.

When a session naturally completes, increment the streak and store the completed goal.

Files to Create

Create this structure:

```
focus-buddy/  
manifest.json  
background.js  
content.js  
character.js  
companion.css  
popup.html  
popup.css  
popup.js  
blocked.html  
blocked.css
```

blocked.js
buddy.webm
scold.mp3
icons/
icon16.png
icon48.png
icon128.png

Manifest

Use Manifest V3.

Required permissions:

```
* 'storage'  
* 'webNavigation'  
* 'tabs'  
* 'alarms'  
* 'scripting'
```

Host permissions:

```
* '<all_urls>'
```

Use 'background.js' as the service worker. Use 'popup.html' as the action popup. Register a content script on '<all_urls>' at 'document_idle' that loads 'character.js', then 'content.js', and injects 'companion.css'. Expose 'blocked.html', 'character.js', 'buddy.webm', and 'scold.mp3' as web-accessible resources for all URLs.

State Model

Store all session state in 'chrome.storage.local'.

Defaults:

```
{  
  active: false,  
  goal: "",  
  endTime: 0,  
  durationMin: 25,  
  allowlist: ["localhost", "127.0.0.1", "youtube.com", "claude.ai"],  
  streak: 0,  
  apiKey: "",  
  lastCompletedGoal: ""  
}
```

Also store captured notes as:

```
capturedNotes: [  
  { text: string, goal: string, timestamp: number }  
]
```

Blocking Rules

Only block 'http' and 'https' URLs. Allow other schemes.

Normalize allowlist entries by:

- * lowercasing,
- * trimming whitespace,
- * removing 'http://' or 'https://',
- * removing path portions after the hostname.

A host is allowed when it exactly equals an allowlist entry or is a subdomain of one. For example, 'mail.google.com' should match 'google.com'.

The background worker should enforce blocking on:

- * tab activation,
- * tab URL update,
- * completed tab load,
- * browser window focus change.

If a content script is not already available in the tab, inject 'companion.css', 'character.js', and 'content.js' using 'chrome.scripting', then send the force-block message.

Background Worker Behavior

Implement:

- * Static fallback line banks for 'encouragement', 'block', and 'complete'.
- * 'get-state', 'start-session', 'end-session', 'get-line', and 'check-blocked' message handlers.
- * 'start-session': set state, calculate 'endTime', create a 'sessionEnd' alarm, create a 'refillLines' alarm every 4 minutes, clear AI line pools, optionally refill AI lines, and greet the active tab.
- * 'end-session(completed)': clear active state and alarms. If completed, increment streak and save 'lastCompletedGoal'.
- * 'sessionEnd' alarm should call 'endSession(true)'.
- * 'refillLines' alarm should generate more AI lines if an API key is present.

For optional AI lines, call the Anthropic Messages API directly from the extension background worker using:

- * endpoint: 'https://api.anthropic.com/v1/messages'
- * model: 'claude-fable-5'
- * 'anthropic-version: 2023-06-01'
- * 'anthropic-dangerous-direct-browser-access: true'

Prompt the model to return JSON only:

```
{
```

```
"encouragement": ["6 short strings"],
"block": ["6 short strings"],
"complete": ["3 short strings"]
}
```

Each line should be under 12 words, lowercase, playful, warm, and a little cheeky. If anything fails, silently fall back to the static line banks.

Content Script Behavior

The content script is responsible for two UI systems:

1. A small corner companion.
2. A full-page blocking overlay.

Guard against duplicate injection with 'window.__focusBuddyLoaded'.

All runtime messaging must go through a safe wrapper that catches invalidated extension contexts. If 'chrome.runtime' is unavailable or 'chrome.runtime.lastError' occurs, fail silently.

On page load:

- * Send 'check-blocked' with 'location.hostname'.
- * If blocked, show the block overlay.
- * If not blocked, show the corner buddy with an encouragement line.
- * If a session is active and the current host is allowed, schedule encouragement popups every random 90 to 180 seconds.

Respond to background messages:

- * 'force-block': show the block overlay.
- * 'buddy-say': show a companion message unless the current page is blocked.

Buddy Media

Use 'buddy.webm' as the visual mascot video. It should be a cute walking/talking girl or pixel-art style buddy animation with transparent or visually removable background. The original asset can be created with tools such as Veo, ChatGPT, Claude, pixel-art tools, animation tools, a walking-girl animation workflow, and background-removal tooling.

Important behavior:

- * The small corner buddy should use 'buddy.webm' muted so autoplay works.
- * The block overlay should show a large version of 'buddy.webm', not looped, sliding in from the right.
- * Use a separate 'scold.mp3' voice clip for the blocked overlay because background removal may remove the original video audio.
- * Trigger 'scold.mp3' about 2.5 seconds after the overlay media is created so it lines up with the talking segment of the video.
- * Attempt autoplay with sound first. If the browser blocks audio, wait for any page

interaction such as pointer movement, pointer down, or keydown, then retry.
* If video playback fails or the extension context is stale, fall back to an inline SVG mascot.

Character Fallback

Create 'character.js' exposing:

```
window.FocusBuddyCharacter.svg(mood)
```

Supported moods:

- * 'cheer'
- * 'stern'
- * 'proud'
- * 'wave'

The SVG should be a cute pink pixel-ish blob mascot with eyes, blush, antenna, arms, and mood-specific mouth/eyebrows/arm pose. The wave mood should include a waving arm that CSS can animate.

Popup UI

The popup should be about 288px wide with a retro Windows/Tahoma look.

Idle view:

- * Titlebar with "focus-buddy" and streak.
- * Mascot SVG.
- * Notes recap from the last session, if any, with a clear button.
- * Goal input.
- * Duration buttons.
- * Collapsible "allowed sites & options".
- * Allowlist textarea.
- * Optional Claude API key password input.
- * Start button.

Active view:

- * Mascot SVG.
- * Current goal.
- * Live countdown.
- * End session button.

Use 'textContent' when rendering user notes to avoid HTML injection.

Overlay UI

The injected overlay should:

- * Use maximum z-index.
- * Freeze document scrolling while active, then restore it when removed.
- * Add a teal blurred scrim with soft pink/yellow radial highlights.
- * Place a retro dialog panel on the left and the large buddy video on the right.
- * Slide the buddy in from off-screen right.
- * Stack the dialog above the character on narrow screens.
- * Respect 'prefers-reduced-motion'.

Dialog content:

- * Title: 'focus-buddy.exe - access blocked'
- * Random block line.
- * 'you're focusing on: <goal>'
- * Live countdown.
- * Buttons: back to work, notes, end session early.
- * Notes textarea and save button when notes panel is open.
- * Streak line.

Alternate Block Page

Also create 'blocked.html', 'blocked.css', and 'blocked.js' as a standalone fallback page. It should use the same retro style, static stern SVG mascot, random block line, goal/from query params, live countdown, back button, end-session button, and streak display. The current main enforcement path may use the injected overlay, but keep this page available as a fallback resource.

Visual Style

Use:

- * Tahoma / Segoe UI / Verdana.
- * Retro desktop-window titlebars.
- * Pink gradient titlebars using '#c0367f' and '#ff5fa2'.
- * Dark purple text/borders '#2a1b3d'.
- * Light gray/lavender panel backgrounds '#dcd7e3' and '#efecf4'.
- * Teal blocked-page background/scrim '#0f8a8a'.
- * Chunky borders and hard offset shadows.
- * Rounded corners around 4-8px.

Acceptance Criteria

- * Loading the unpacked extension shows the popup.
- * Starting a session stores state and changes the popup into active countdown mode.
- * Allowed sites remain usable.
- * Disallowed sites get the injected overlay.
- * The overlay countdown updates every second.
- * The "back to work" button leaves the distracting page.
- * Notes can be saved from the overlay and later appear in the popup recap.
- * Ending early clears the session without incrementing streak.
- * Natural completion increments streak.

- * The buddy appears on allowed pages during a session with encouragement every 90-180 seconds.
- * The block overlay plays the buddy video and attempts the scold voice around 2.5 seconds after appearing.
- * If media or extension context fails, the app degrades gracefully to the SVG buddy and static line bank.